Energy Reports 9 (2023) 1113–1123

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

A multi-objective optimal PMU placement considering fault-location topological observability of lengthy lines: A case study in OMAN grid

Amer Al-Hinai [a,b,*,1], AliReza Karami-Horestani [b,1], Hassan Haes Alhelou [a,c,*,1]

[a] Department of Electrical and Computer Engineering, Sultan Qaboos University, P.O. Box 33, Muscat 123, Oman
[b] Sustainable Energy Research Center, Sultan Qaboos University, Muscat, Oman
[c] Department of Electrical and Computer Systems Engineering, Monash University, Clayton VIC 3800, Australia

ARTICLE INFO

Article history:
Received 22 August 2022
Received in revised form 29 November 2022
Accepted 16 December 2022
Available online 23 December 2022

Keywords:
Phasor measurement unit
Optimal PMU placement
OMAN grid
Fault location observability (FLO) index
MILP-based multi-objective optimization

ABSTRACT

Using PMU data, the fault location algorithm has gradually been trended in the literature. Besides the optimality of the algorithm, the fault location application should directly receive the data of the faulty line from the PMU devices to have the best performance; using state estimation to create the input data of fault location algorithm not only wastes the golden time, but it also decreases the reliability. Given, it is not possible to install PMU devices in all buses of the system, it is better to install them for the lengthy lines having the most probability of the failure. Although various approaches have been presented in the literature to solve optimal PMU placement (OPP), the fault location observability has less been considered in OPP formulation Accordingly, this paper considers a new index, named fault location observability (FLO), as a second objective function in its OPP formulation. Using this index, in addition to minimizing PMU installation cost, the number of lengthy lines being directly observable are also maximized. Using the ε-constraint method, the multi-objective problem has been solved several times as single-objective optimizations. Given the constraints and objective functions are linear, the work presents a mixed integer linear programming (MILP) formulation for the placement problem. Using the GAMS under CPLEX solver, the global optimal solutions of MILP model corresponding to the various ε-constraints yields to a Pareto optimal front. The simulation results, next, implements the proposed method on the IEEE 300 bus test system as well as the practical OMAN grid to prove the effectiveness of the proposed method.

1. Introduction

Phasor measurement units (PMUs) provide a time-synchronized accurate measurement in power system networks (Zhang et al., 2022; Ahmed et al., 2022). In the near future, the present-day SCADA system will completely be replaced with PMUs. Using the global positioning satellite (GPS) system, the PMUs have improved the accuracy of measurement. Consequently, monitoring and controlling of the power systems are becoming directly dependent on their PMUs (Lin et al., 2022). These measuring units play a vital role in the future power systems; using the advanced communication capabilities in collecting, monitoring and archiving the real-time data, the PMUs have the ability to make the power system fully observable even in dynamic analysis (Appasani et al., 2022; Jha et al., 2022).

Although installing PMUs in all nodes of a sample network makes it fully observable, the following three reasons persuade any individual to make the network topological observable with the least number of PMUs. That is, (Ahmed et al., 2022; Baba et al., 2021):

- Installing PMUs in all buses of a network is a *costly* solution for full observability,
- The KVL and KCL laws let the researchers make some buses of the network *indirectly* observable without installing individual PMUs,
- Installing numerous PMUs in the network yields the system to the *big-data challenge* in which the system designer should find a way for omitting the extra data, prioritizing the PMUs in storing information, verifying, sampling, archiving and … the data.

Although a network may be topologically observable, while it is not really observable (because of the gain matrix singularity, the network at that working point is numerically non-observable). According to the rarity of this phenomenon, however, this paper

* Corresponding authors at: Department of Electrical and Computer Engineering, Sultan Qaboos University, P.O. Box 33, Muscat 123, Oman.
E-mail addresses: hinai@squ.edu.om (A. Al-Hinai), hassan.haesalhelou@monash.edu (H.H. Alhelou).

[1] The authors confirm that they contributed to this work equally.

https://doi.org/10.1016/j.egyr.2022.12.046

checks the observability of the network only from topological point of view (Qi et al., 2015; Johnson and Moger, 2021).

Accordingly, assigning the optimal number of PMUs as well as allocating them in the network are two important engineering decisions in managing the modern power system (Babu et al., 2022). In the last two decades, accordingly, the Optimal PMU Placement (OPP) has gradually become a hot trend among the researchers in the literature (Ahmed et al., 2022; Dixit et al., 2022). The main objective function of these approaches is usually to determine the minimal number (or cost) of PMUs in the network (Mishra and Cherukuri, 2020).

In addition to cost minimization, the PMU placement problem has also the ability to optimize a second objective function. Given the PMUs would usually perform various duties and applications in the operation of a practical network, it is possible to reallocate the PMU locations to optimize the performance of one of these applications. Among them, the fault location algorithm is one the most important one. That is, the electrical faults in the lines are probable in power systems and the system designers are doing their best to protect the system against them (Alexopoulos et al., 2016; Theodorakatos, 2019). One of the most important portions of the protection system in repairing the system and returning the line back into service is estimating the location of the occurred fault (Hamedani Golshan et al., 2018). On the other hand, the modern power systems are usually being operating close to their operational limits. In such a condition, the location of fault should be found and cleared as soon as possible. Accordingly, the future power systems should be equipped with a fast and confident fault location plan.

Various researchers have proposed different fault location methods which can be classified into two following categories:

- Traveling wave (TW) oriented methods: these methods detect the fault location according to TW theory (Naidu and Pradhan, 2021). Although these approaches can effectively locate the fault point, the researchers should find a practical solution for the main challenges of this methodology, i.e. the costly hardware for detection of TWs, multiple reflections, noise, detection of wave front, detecting high impedance fault.
- Phasor measurement oriented methods: these approaches use the current and voltage phasors at the ends of transmission lines to detect the fault location. Some of these methods require the data of the both sides of the line, while some others require only one-side data (Jia et al., 2013).
  Using relays or digital fault recorders (DFR), it is possible to calculate the voltage and current phasors from the gathered time series data via phasor estimation algorithms (Galvez and Abur, 2022; Joshi and Verma, 2021). Moreover, for the two-sided fault location algorithms, it is also necessary to find a way for synchronizing the time series data. That is while, the PMUs calculates the phasors without any need to phasor estimation algorithms or synchronization methods.

With the invention of PMU devices, the literature has seen various PMU-based algorithms in estimating the fault location. Every day, these algorithms are growing in the literature and of course, one of them may be the best choice for the desired practical network. However, before applying it on the network, it is necessary to have a plan for its requirements; the fault location algorithms usually need the PMU data at the nodes of the line containing the fault. That is, the indirect observability of a node via an application like state estimation or any similar approaches is not desirable during the fault condition. Dependency of the fault location algorithm to state estimation as well as communication structure or CPU processes not only decreases the reliability of the method, but it also wastes the golden time of fault location process. Thus, it shall be necessary to install PMU devices at the nodes connected to the lines with the most probability of the failure.

Table 1 presents a review on the OPP papers in the literature and compares their characteristics with the proposed method.

As shown in Table 1, Ref. Babu et al. (2022) uses a nonlinear programming (NLP) based approach to solve the OPP problem, not only during the normal operating condition, but even also during the contingencies of PMUs or lines. Moreover, a voltage stability proximity index is also utilized to find out the location of the line loss in the network. Using a graph neural network, Lv et al. (2022) proposed an accurate distribution network state estimation model which requires only a low-resolution measurement data. Next, this paper presents an OPP algorithm which uses genetic algorithm (GA) to find out the best places and the number of PMUs guaranteeing the proposed state estimation data accuracy. The authors in Hyacinth and Gomathi (2021) utilized a three-stage algorithm to determine, first, the minimum number of PMUs making the system fully observable. Second, the minimum installation cost and third, the maximum improvement in measurement redundancy. A new method for solving OPP using the integer linear programming (ILP) is presented in Elimam et al. (2021). The proposed method considers the parameters of different branches in the formulation of the optimization problem. In this way, the network is decomposed into smaller sub-networks with reduced dependence among them making it possible to scale them when applied to typical networks. The authors in Andreoni et al. (2021) proposed a multi-objective PMU placement strategy for minimizing the total number of PMU channels, the uncertainty in state estimation and the sensitivity of the state estimation to the parameters of the branches. Using the Binary Dragonfly Algorithm (BDA), The simultaneous optimization of PMU installation and communication infrastructure costs are done in Patel et al. (2022). In this work, the effect of zero injection buss, N-1 contingencies, pre-installed fiber optics and also pre-installed PMUs are also considered in simulation. The authors in Cao et al. (2022) discusses the incomplete observability under PMU loss. The five-objective placement problem of this paper is solved via an improved two-archive algorithm. Moreover, a fuzzy decision-making approach is also presented in this paper helping the system designers select the best solution for their work. The limitation caused by zero injection buses as well as conventional measurements in PMU placement are well considered in Khajeh et al. (2017). The minimum number of PMUs, maximum measurement redundancy, N-1 contingencies and the effect of conventional non-synchronous measurements were well considered in the multi-objective OPP formulation of Mazhari et al. (2013). This paper used Binary PSO for finding its best optimal solutions. Using the mixed integer quadratically constrained programming (MIQCP), The authors of Zhu et al. (2019) extended the OPP formulation to take into consideration both power system observability and data transmission bandwidth requirements. Using binary genetic algorithm (BGA), a multi-stage OPP formulation is presented in Almasabi and Mitra (2018). In this paper, the PMUs were installed over the span of years according to the budget limitation.

According to Table 1, enhancing the fault location applications has been less considered in OPP formulations; the main reason may be related to the fact that fault location algorithms are usually been noticed during operating phase studies and they are not assumed to be important during planning phase. Accordingly, this paper defines a new objective function focused on enhancing fault location requirements for the OPP problem; this objective function makes the optimization problem install the PMU devices in the nodes connecting to the lines with the most probability of the failure (lengthy lines). In this way, for the majority of failures, the future power system will use fault location algorithms

Table 1
The comparison of PMU placement papers with the proposed method.

Ref.	Technique	Objective functions		Constraints			
		Minimize PMU ...	Optimize fault location needs	Observ.	N-1 Contingency	Channel Limitation	Zero inj.
Babu et al. (2022)	NLP using GA	Number	–	✔	✔	–	✔
Lv et al. (2022)	Graph Neural Network	Number	–	✔	–	–	–
Hyacinth and Gomathi (2021)	A Heuristic Three-Stage Method	Number	–	✔	–	–	–
Elimam et al. (2021)	ILP using MATLAB optimtool	Number	–	✔	✔	–	✔
Andreoni et al. (2021)	NSGA II	Cost	–	✔	✔	✔	✔
Patel et al. (2022)	BDA Algorithm	Number	–	✔	✔	–	✔
Cao et al. (2022)	MILP	Cost	–	✔	✔	✔	✔
Khajeh et al. (2017)	MILP using GAMS	Number	–	✔	✔	–	✔
Mazhari et al. (2013)	Binary PSO	Number	–	✔	✔	✔	✔
Zhu et al. (2019)	MIQCP	Cost	–	✔	✔	✔	✔
Almasabi and Mitra (2018)	BGA	Cost	–	✔	–	✔	✔
Proposed	MILP using GAMS	Cost	✔	✔	✔	✔	✔

without any need to performing state estimation on the PMU data.

Hence, the OPP formulation should be changed in a way that it installs the PMUs near the longest lines. Accordingly, defining a length-oriented index for the fault location observability of the system, this paper considers this subject as a second objective function in a multi-objective formulation. In other words, in addition to buses, this paper is going to maximize the number of fault-location observable lines from a predefined candidate set (lengthy lines), while simultaneously minimize the PMU installation cost in the network.

The salient features of this paper are as follows.

- A new objective function, named fault location observability (FLO), presented showing the capability of the network in better utilizing the future PMU-based fault location applications.
- A multi-objective framework is presented to minimize both PMU installation costs and fault location observability.
- The proposed algorithm equips the power system with fault location applications without any extra investment for DFRs/relays or any data synchronization requirement.
- According to the linearity of equations, the problem is stated as a mixed integer linear programming (MILP) model in the general algebraic modelling language (GAMS) software under CPLEX solver which guarantees the global optimality of the final solution.
- The ε-constraint method is used to generate the optimal Pareto front of the MILP problem.
- An index, named FLO/PMU is presented to help the decision-maker to affordably select its solution among the Pareto optimal front.
- The effect of N-1 contingencies on the final results of OPP are well considered.

Accordingly, the rest of the paper is organized as follows.

Section 2 answers "what is a PMU device?". Section 3 defines what is the node observability in the language of mathematics. This section also presents the theory of zero-injection nodes and its impact on node observability. Defining the cost function of PMU placement as well as an objective function to maximize the FLO lines according to their length, Section 4 presents the multi-objective formulation of this paper. Section 5 applies the formulation on the practical OMAN network within two case studies and for normal operation and N-1 contingencies. In addition, to prove the scalability of the methodology the proposed method is also applied on IEEE 300 bus test system. Finally, Section 6 concludes the paper.

2. What is a phasor measurement unit?

Using a time source for synchronization, PMU estimates the phase angle and magnitude of voltage and current in a power grid. PMUs make the power system capable of real-time measurements of multiple remote points. These devices can capture high-frequency samples of voltage and current waveforms, making them capable of reconstructing the phasor quantity of voltage or current (Mishra and Cherukuri, 2020) (see Fig. 1).

A PMU has generally four following components:

(1) GPS Receiver: this component mainly creates a time tag for the measured signal.
(2) Microprocessor: Considering the GPS data at every initial stage, the microprocessor identifies the number of samples. Moreover, using a recursive algorithm, this component has the ability to control the phasors.
(3) Phase-Locked Oscillator: to reach a sampling rate of 30–48 cycle per second (the normal rate for a standard PMU), this component divides the 1 pps into required number of pulses per second.
(4) Surge Filtering: having a similar attenuation and phase shift in the transformer output (voltage or current) is what this component does.

3. Node observability

Assuming a PMU has the ability to measure the respected node voltage phasor and also the current phasors of all connected branches, a node is observable when

- A PMU is installed on that node, or
- A PMU bus is directly connected to it.

Accordingly, for any set of PMU installation in different nodes of the system, it is possible to calculate the observability as follows:

$$x_i = \begin{cases} 1, & \textit{if there is PMU in node } i \\ 0, & \textit{if there is no PMU in node } i \end{cases} \tag{1}$$

Where x is a binary variable denoting whether a PMU is installed at bus i or not.

Using the node incidence matrix, A, of the network, in which the ijth component is defined as follows (Eq. (2)), it is possible to define the observability of a specific node as Eq. (3).

$$a_{ij} = \begin{cases} 1, & \textit{if } i = j, \\ 1, & \textit{if } i \textit{ and } j \textit{ nodes are connected}, \\ 0, & \textit{if there is no PMU in node } i \end{cases} \tag{2}$$

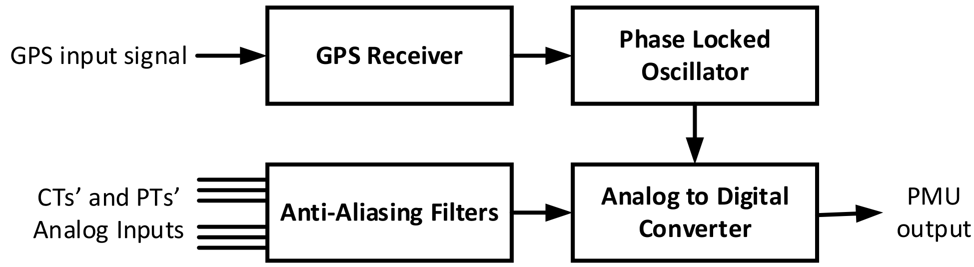

Fig. 1. The main common component of a PMU device.

$$u_i = \sum_{j=1}^{N} a_{ij} m_{ij} \tag{3}$$

Where u_i denotes the observability of node i, N is the total number of nodes in the network, a_{ij} is the ijth component of A matrix and m_{ij} is a binary variable presenting whether the PMU in bus i has a current channel in the corresponding ijth branch of the network or not. Eqs. (4) to (6) show the mathematical presentation of this variable.

$$m_{ij} \le x_i \tag{4}$$

$$m_{ij} \le a_{ij} \tag{5}$$

$$m_{ii} = x_i \tag{6}$$

Eq. (4) shows that if x_i was zero, consequently the m_{ij} is always zero, because there is no PMU at bus i to have a current channel in the ijth branch. However, if there is a PMU at bus i, the m_{ij} may be zero, because the respected PMU may have no channel at one of its branches. Eq. (5) denotes an obvious fact that m_{ij} can be non-zero if there is a physical branch between bus i and j ($a_{ij} = 1$). Eq. (6) denotes that each PMU has a voltage channel for measuring the voltage magnitude at its own node. According to the above equations, a network is fully observable when all of its nodes are observable ($u_i > 0$ for all nodes of the system).

Given the zero-injection nodes have the ability to extend the observability through the network without installing a new PMU, it is also necessary to consider this fact in the equations of observability; a zero-injection node injects no current to the power system. There is no need to install any PMU at zero-injection nodes; the Kirchhoff current law (KCL) at such kind of buses provides no new information for the network engineer. Accordingly, considering this issue, the observability function of Eq. (3) will be redefined as follows.

Observability function

$$u_i = \sum_{j=1}^{N} a_{ij} m_{ij} + \sum_{j=1}^{N} a_{ij} y_{ij} \ge 1 \tag{7}$$

$$z_i = \sum_{j=1}^{N} a_{ij} y_{ij} \tag{8}$$

Where, z_j is a binary parameter showing whether bus j is a zero-injection bus or not. y_{ij} is also an auxiliary Boolean variable, related to buses i and j, showing whether bus i is observable by the effect of zero-injection bus at bus j or not. It is obvious that the zero-injection buses have at least one non-zero auxiliary variable.

4. The proposed opp formulation

This section presents the multi-objective formulation for the simultaneous optimization of PMU placement cost function and fault location observability of the network lines. Accordingly, the formulation of these two objective functions is presented respectively.

4.1. PMU placement cost function

In the basic OPP problem, finding the minimum number of PMUs is the main objective function making the network fully observable. The mathematical representation of such a famous objective function is given below:

$$\min f_1 = \sum_{i=1}^{N} x_i \tag{9}$$

However, in the reality the cost of installing PMU in different buses is not identical; the PMU should measure the voltage and current phasors at all the connected branches of the respected bus. Accordingly, a number of CT and PT devices should also be installed corresponding to the outgoing branches of that busbar. That is, the placement cost is dependant to the number of outgoing branches in that bus, which is proportional to the number of PMU channels.

Considering the PMU cost as a function of its channel numbers, the PMU placement cost function can be rewritten as follows:

$$\min f_1 = f_{cost} = \sum_{i=1}^{N} c_F x_i + \sum_{i=1}^{N} \sum_{j=1}^{N} c_{ch} m_{ij} \tag{10}$$

In this equation, c_F is the fixed investment cost and c_{ch} is the variable investment cost corresponding to a measurement channel and m_{ij} is a binary variable showing whether the PMU installed at bus i has a channel to jth bus through the connected line or not.

4.2. Fault location observability objective function

As the lengthy lines usually have a larger probability to experience a storm, lightning, snow, birds' accident or insulation breakdown (Hamedani Golshan et al., 2018), to quantify this fact in the multi-objective formulation, the following objective function is defined.

$$\min f_2 = f_{FLO} = \sum_{i=1}^{N} \sum_{j=1}^{N} l_{ij} \left(1 - m_{ij}\right) \tag{11}$$

Where f_{FLO} is the FLO objective function and l_{ij} is the length of the line between ith and jth nodes.

According to this equation, f_{FLO} is the summation of length in the non-FLO lines. Minimizing this objective function will automatically increase the importance of installing PMUs at the lines with greater length.

4.3. Optimization constraints

In addition to Eqs. (4) to (6), the following equations should also be considered in this formulation to obtain a correct set of solution for this problem.

$$\sum_{j=1}^{N} a_{ij} m_{ij} \le Ch_{max} \tag{12}$$

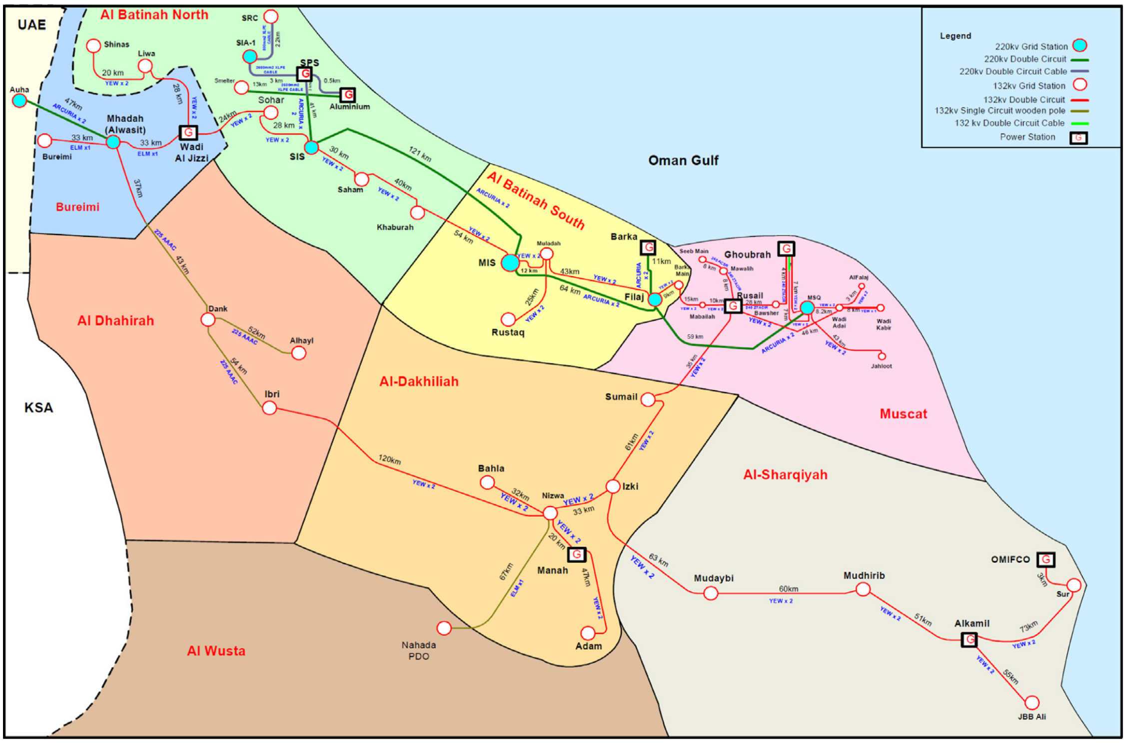

Fig. 2. The geo-schematic diagram of OMAN transmission grid (Abdalla et al., 2009).

$$u_i \geq 1 \qquad \forall i \epsilon \{1 \ldots N\} \tag{13}$$

Where Ch_{max} is the maximum allowable number of PMU channels in a node and Eq. (12) considers the constraint of maximum channel for a PMU device. Similarly, Eq. (13) denotes the necessity to have observability at all nodes of the system.

For considering the N-1 contingencies in the formulation, Eq. (13) is replaced with Eq. (14).

$$u_i \geq 2 \qquad \forall i \epsilon \{1 \ldots N\} \tag{14}$$

This equation means that each bus of the system should become observable from two separate directions. In this way, in the occurrence of an N-1 contingency the system is yet observable.

5. Simulation results

In this part, two case studies are considered for simulation. One of the is the famous IEEE 300 bus test system and the second is OMAN practical transmission grid.

As shown in previous section, all the equations of the proposed method are linear and because of that it is possible to define this formulation as a MILP problem. Accordingly, this section has implemented the proposed MILP formulation within the GAMS software under the famous CPLEX solver to surely reach the global optimum. All the simulation tests in this section are carried out on a 2.4 GHz Intel (R) Core (TM) i5-2430M based desktop.

The Optimal PMU placement algorithm has been validated on the practical OMAN transmission network which extends across the whole northern OMAN and connects the generators to the respected loads located in the governorate of Muscat and the regions of Batinah, Dhahirah, Dakhliyah and Sharquiya (Abdalla et al., 2009). Fig. 2 shows a geo-schematic diagram of the OMAN transmission grid. This network has 132 buses and 172 branches. The following table illustrates the characteristics of the system lines.

As shown in Table 2, this system has numerous numbers of lines with different lengths. These lines are connecting the "from" node number to the respected "to" node number.

In this section, three case studies are considered for the studied system, the first one minimizes the number of PMUs as a single-objective function, the second presents the Pareto optimal solutions of the multi-objective problem and the third case study takes the N-1 contingencies into consideration.

5.1. Single-objective optimization: minimizing PMU placement cost

The cost data used in this simulation are taken from (Arpanahi et al., 2019) as $c_F = 20000$ \$ and $c_{ch} = 1000$ \$. The maximum number of channels per PMU is also assumed to be 8 ($Ch_{max} = 8$).

Considering the 36 zero-injection busbars, the optimization model, which is implemented in GAMS software, finds the best PMU number and places within these 132 busbars as follows:

In this case study, 28 PMUs with the total number of 96 channels are recommended to be installed at the buses shown in Table 3. That is, this network will become affordably observable with 28 PMUs with the channels and placements mentioned in the table.

Fig. 3 shows the detailed number of channels which is required to be used for each PMU device in the OMAN network.

5.2. Multi-objective optimization: minimizing both $f_{\mathbf{cost}}$ and $f_{\mathbf{FLO}}$ (normal operation)

In this case study, the optimization problem is going to minimize both objective functions; the PMU placement cost and the fault location observability objective function. The Table 4 shows the optimal Pareto solution.

As shown, from the 10 selected solutions, the first and last answers are the limit states in which the first and second objective functions are optimized by their own, respectively. That is, the first solution is the same as the results obtained in the first case study. As shown, even for this case study, 68 branches of the network become FLO; these two objective functions are not completely conflicting each other and focusing on one of them will unconsciously improve the other one. However, what is different

Table 2
The characteristics of lines in the network.

Name	From	To	Length	Name	From	To	Length	Name	From	To	Length	Name	From	To	Length
L01	131	124	195	**L37**	119	116	37	**L73**	76	46	24	**L109**	1	75	7
L02	125	43	191	**L38**	113	114	37	**L74**	34	11	23	**L110**	41	86	7
L03	126	131	167	**L39**	47	19	37	**L75**	73	72	23	**L111**	27	18	7
L04	120	126	160	**L40**	16	10	37	**L76**	29	66	22	**L112**	27	18	7
L05	125	128	152	**L41**	74	55	36	**L77**	129	132	22	**L113**	26	27	7
L06	43	124	129	**L42**	90	47	36	**L78**	65	48	20	**L114**	50	69	6
L07	130	132	120	**L43**	94	22	36	**L79**	40	76	20	**L115**	5	15	6
L08	127	51	120	**L44**	83	12	35	**L80**	88	78	19	**L116**	69	39	6
L09	110	104	119	**L45**	58	44	35	**L81**	86	38	19	**L117**	66	56	6
L10	101	51	117	**L46**	23	73	35	**L82**	63	31	17	**L118**	118	119	6
L11	28	85	111	**L47**	89	52	34	**L83**	72	88	17	**L119**	70	84	10
L12	101	132	106	**L48**	98	106	34	**L84**	7	94	17	**L120**	42	62	4
L13	85	57	101	**L49**	116	118	33	**L85**	31	17	17	**L121**	3	26	4
L14	35	30	99	**L50**	97	102	33	**L86**	65	17	16	**L122**	6	89	4
L15	120	123	85	**L51**	84	87	33	**L87**	96	111	16	**L123**	68	21	4
L16	129	131	81	**L52**	47	20	33	**L88**	16	56	15	**L124**	119	115	4
L17	111	106	76	**L53**	65	13	32	**L89**	82	81	14	**L125**	9	69	4
L18	65	60	67	**L54**	91	64	32	**L90**	9	53	14	**L126**	9	69	4
L19	116	110	67	**L55**	47	93	32	**L91**	122	121	13	**L127**	59	6	3
L20	110	118	62	**L56**	8	33	30	**L92**	65	45	12	**L128**	4	59	3
L21	107	116	61	**L57**	76	79	30	**L93**	5	44	12	**L129**	105	104	3
L22	33	83	61	**L58**	55	77	29	**L94**	41	58	12	**L130**	54	53	3
L23	31	81	61	**L59**	9	82	29	**L95**	123	130	11	**L131**	4	89	3
L24	56	55	60	**L60**	78	90	28	**L96**	70	84	10	**L132**	18	42	3
L25	93	22	58	**L61**	30	24	28	**L97**	15	54	10	**L133**	40	79	2
L26	22	30	56	**L62**	64	58	28	**L98**	97	100	10	**L134**	71	80	2
L27	111	100	55	**L63**	38	37	28	**L99**	42	67	10	**L135**	117	119	2
L28	32	10	55	**L64**	104	103	28	**L100**	97	112	10	**L136**	18	62	2
L29	61	91	55	**L65**	90	79	27	**L101**	37	36	9	**L137**	6	92	3
L30	110	99	51	**L66**	109	111	26	**L102**	3	50	9	**L138**	39	50	2
L31	77	63	51	**L67**	96	112	25	**L103**	1	53	9	**L139**	92	59	3
L32	48	2	49	**L68**	34	95	25	**L104**	3	49	9	**L140**	100	97	5
L33	68	34	49	**L69**	42	11	25	**L105**	42	89	8	**L141**	20	19	6
L34	108	107	46	**L70**	36	23	24	**L106**	50	75	8				
L35	108	102	46	**L71**	65	35	24	**L107**	67	6	8				
L36	10	33	44	**L72**	5	61	24	**L108**	1	75	7				

Table 3
The placement results in the first case study.

# of buses	# of zero-injection buses	# of optimal PMUs	# of optimal channels	f_1 ($)	PMU placements
132	36	28	96	656000	3, 7, 8, 10, 18, 30, 31, 34, 37, 38, 47, 48, 50, 52, 53, 55, 58, 59, 61, 65, 68, 76, 80, 83, 85, 88, 119, 125

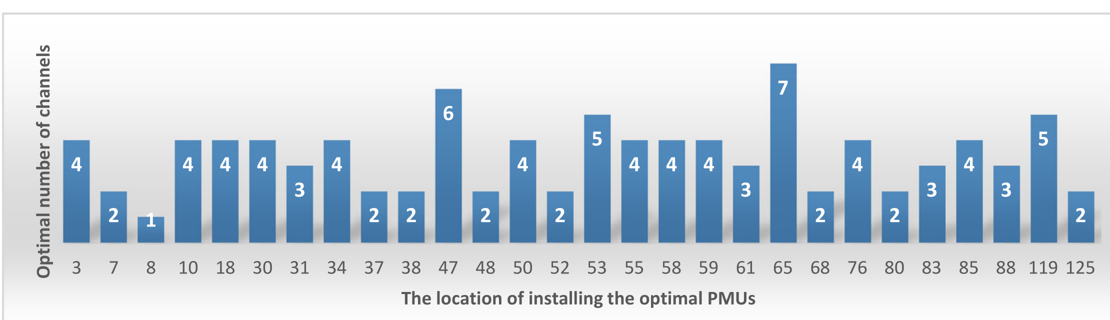

Fig. 3. The detailed number of channels in the optimal solution of first case study.

is the point where each of these objective functions experience their minimum. That is the exact reason that decision-maker needs to perform a trade-off among the solutions.

In this table, the second column shows the criterion used in ε-constraint methodology to select the final solution from the optimal Pareto set. As shown, if the decision-maker decided to have a specific value as the maximum for f_{FLO}, he/she can easily find its best solution.

The third column denotes the first objective function value corresponding to the selected solution. It is worth mentioning that, for the sake of simplicity, the values are illustrated in the unit of k$ which is equal to 1000 $.

The fourth column shows the value of the second objective function. As mentioned in Section 5, this value is the summation of the length of the lines which are not assumed FLO. Thus, minimizing this value will increase the number of FLO lines in the network.

The fifth column denotes the number of PMUs suggested to be installed in the corresponding solution. As shown, the first solution has 28 PMUs while the last solution suggests to install

Table 4
The pareto optimal solution in the second case study.

	Criterion	f_{cost} (k$)	f_{FLO} (km)	# of PMUs	# of channels	# of FLO lines
Solution 1	Only f_{cost}	656	8298	28	96	68
Solution 2	f_2 < 7000	683	6980	29	103	74
Solution 3	f_2 < 6500	712	6494	30	112	82
Solution 4	f_2 < 6000	752	5994	32	112	80
Solution 5	**f_2 < 5000**	**836**	**4999**	**35**	**136**	**101**
Solution 6	f_2 < 4000	965	3995	41	145	104
Solution 7	f_2 < 3000	1159	2999	49	179	130
Solution 8	f_2 < 2000	1432	1999	61	212	151
Solution 9	f_2 < 1000	1911	998	82	271	189
Solution 10	Only f_{FLO}	3106	0	132	466	334

Table 5
The FLO/PMU index for 10 solutions.

	S #1	S #2	S #3	S #4	S #5	S #6	S #7	S #8	S #9	S #10
FLO/PMU	2.43	2.55	2.73	2.50	**2.89**	2.54	2.65	2.48	2.31	2.53

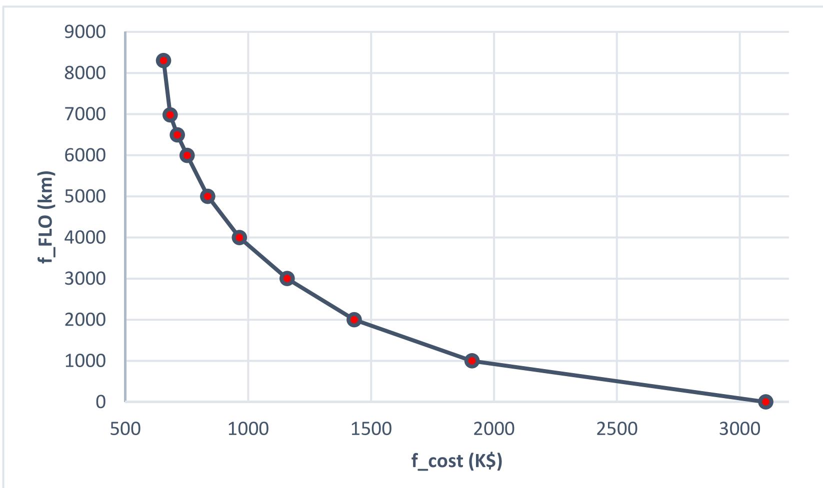

Fig. 4. The illustration of Pareto optimal front.

132 PMUs at all busbars. This issue also illustrates the difference between these two limit cases. Minimizing the PMU placement cost yields to the least possible number of PMUs (28), while minimizing the non-FLO branches lead to installing PMU at all busbars without any concern on the placement cost.

The sixth column shows the total number of channels used in the installed PMUs in that solution. As described in Section 4, in addition to the own PMU, the number of its channel has a significant impact on the cost of PMU placement. Accordingly, as shown in this table, the number of channels increases in this column from up to down.

The seventh column illustrates the number of FLO lines in the respected solution. Given each line has two ending points, the FLO index is tested from two sides and as a result the number of FLO lines in this column is twice the number of FLO lines. However, sometimes a line may assume FLO from one side and non-FLO from the opposite side. Thus, this value may be odd in different solutions.

Fig. 4 illustrates the Pareto optimal front.

As shown in Fig. 4, there is a trade-off between the two objective functions; none of the them can be improved without degrading the other objective value. That is, if the decision-maker decided to increase the number of FLO lines, he should accept a little increase in the cost of PMU placement.

5.3. Selecting the most affordable solution among pareto front

Although, it is rational to delegate selecting the best solution to decision-maker, if there was an index for optimal selection among the solution, he/she would better perform this responsibility. Accordingly, the following figure will help him/her in deciding the best answer.

As shown in Fig. 5, increasing the number of PMUs has the largest effect in solution 5. In other words, the number of FLO lines per PMU has the largest value in this solution. In addition to the above figure, it is also possible to define the following index, named FLO/PMU.

$$FLO/PMU = \frac{the\ number\ of\ FLO\ lines}{the\ numebr\ of\ PMUs} \tag{15}$$

The following Table 5 shows the value of this index at different solutions.

As shown, in this table the fifth solution has the best FLO/PMU index among the others and this solution has better affordability. However, it is noteworthy that this index and the corresponding figure cannot replace the responsibility of decision-maker in deciding which solution is his/her choice; the decision-maker has the authority to estimate the value of each objective function in different points of the Pareto front. In other words, this index can help him in deciding among two or three solutions which are completely near each other.

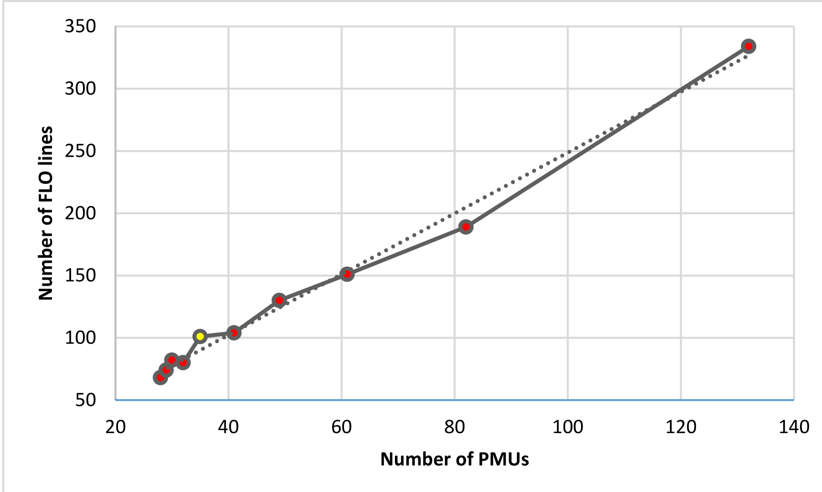

Fig. 5. The illustration of the number of FLO lines with respect to the number of PMUs.

Table 6
The placement results of the fifth solution selected among other results according to FLO/PMU index.

# of FLO lines	# of optimal PMUs	# of optimal channels	f_{FLO} (km)	f_1 (k$)	PMU placements
101	35	136	4999	836	3, 6, 10, 18, 30, 31, 33, 34, 38, 43, 47, 48, 50, 53, 55, 58, 61, 65, 68, 71, 76, 78, 83, 85, 89, 94, 110, 116, 119, 120, 124, 125, 126, 131, 132

Table 7
The placement results of the fifth solution selected among other results according to FLO/PMU index.

# of FLO lines	# of optimal PMUs	# of optimal channels	f_{FLO} (km)	f_1 (k$)	PMU placements
157	71	228	4902	1648	1, 2, 3, 5, 6, 7, 8, 10, 12, 13, 15, 18, 19, 21, 22, 24, 27, 28, 30, 31, 32, 33, 34, 36, 37, 41, 42, 45, 46, 47, 48, 49, 50, 51, 53, 55, 56, 57, 59, 63, 64, 65, 66, 68, 69, 71, 73, 74, 76, 78, 79, 80, 82, 83, 84, 85, 86, 88, 89, 91, 94, 95, 104, 110, 111, 117, 119, 124, 125, 128, 131.

Assuming decision-maker selects the fifth solution, the respected detailed result is presented in Table 6.

As shown in this table, to have a better number of FLO lines in the network, in this solution the number of PMUs increases to 35 (7 PMUs more than the 1st case). As a result, the total cost of PMU placement has increased to almost 836000 dollars (about 30% more than the 1st case). On the other hand, instead of this increase in PMU installation cost, this solution has increased the FLO lines to 101 lines which is about 50% more than the first case study.

In addition to the PMUs' location, this solution also presents which lines are becoming FLO. Accordingly, Fig. 6 illustrates the FLO lines of the network.

As shown in this figure, some of the lines have an FLO level of 2. These lines are the ones which are becoming observable twice (from both sides), while the others become observable once and only from one side. Given the double-ended fault location algorithms are more accurate than single-ended ones, making the branches observable from both sides is the ideal solution from the second objective function viewpoint. That is why, the tenth solution, in which there is no constraint on PMU placement cost, the number of FLO branches becomes 334. That is, the tenth solution makes almost all the branches of the network observable from both sides.

However, in the fifth solution, the trade-off between the cost and FLO functions yields to have 88 FLO branches. As shown, the two-sided FLO branches are almost placed in the first part of figure containing the largest lines of the network. Meanwhile, the node observability constraint as well as cost function consideration lead the solution to have less two-sided FLO lines in the first part and have some rare two-sided ones in the second and third part.

5.4. Multi-objective optimization: minimizing both $f_{\mathbf{cost}}$ and $f_{\mathbf{FLO}}$ (considering N-1 contingencies)

In case of PMU loss or branch outage, the observability of the system may be threatened. Accordingly, this case study has performed the N-1 study for the best solution of the network to verify the impact of this issue on the final results.

Table 7 shows the results of fifth solutions when the N-1 constraint is also taken into consideration.

As shown in this table, the N-1 contingencies have increased the number of PMUs to 71 (2*35). That is, considering N-1 have doubled the PMUs, the total cost and the number of channels.

5.5. Comparison with other methods and scalability study

To evaluate the scalability of the proposed method, the IEEE 300 bus test system is utilized to verify the method on a larger network. To do so, similar to previous simulations, this case study is also carried out on a 2.4 GHz Intel (R) Core (TM) i5-2430M based desktop. Given the simulation time of OMAN network is in the order of 0.05 s and the 300-bus test system last about 0.1 s, it is possible to conclude that the simulation time is not a vital issue in this formulation. Moreover, it is necessary to point out that this formulation is a linear MILP problem which can be easily converged within the GAMS software; in the OPP problems, which is assumed a kind of planning study, the timing issues are not a critical concern for the researchers.

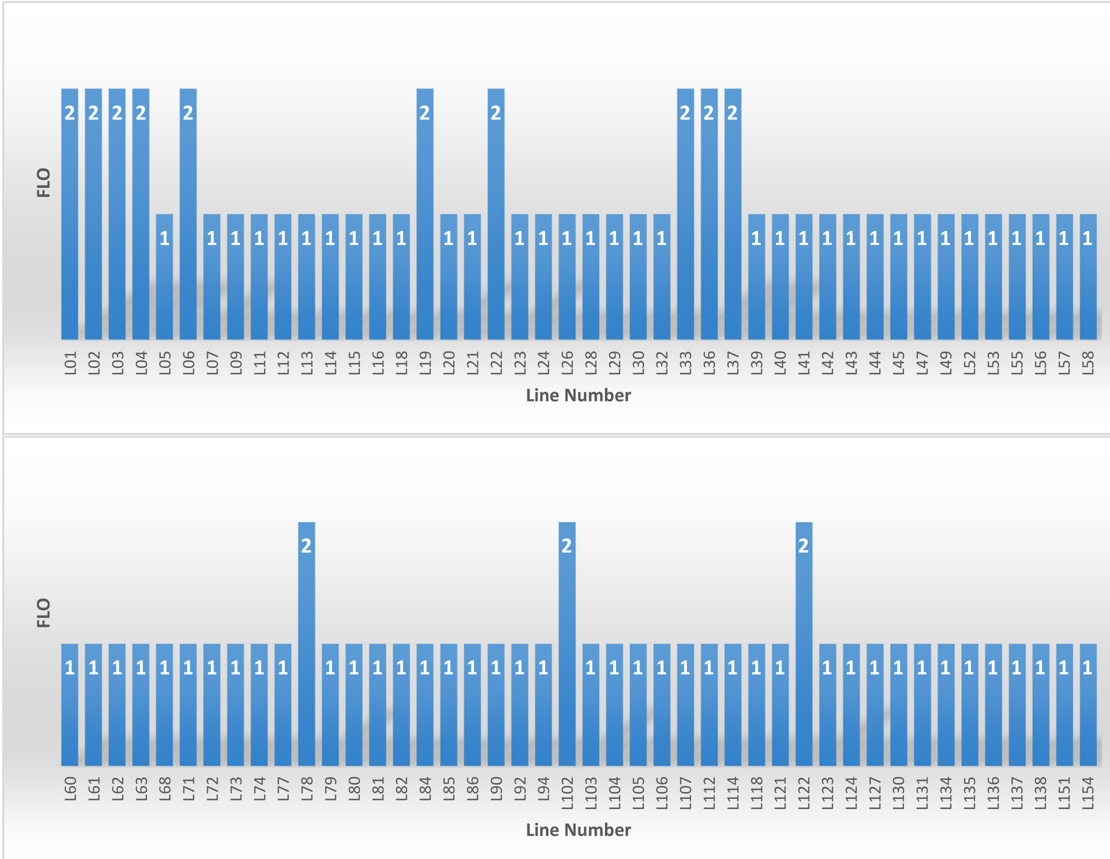

Fig. 6. The FLO lines of the network in the fifth solution.

Table 8
The comparison among the proposed method and the OPP papers working on IEEE 300 bus test system.

Ref.	Method	# of PMUs	Considering FLO
Mazhari et al. (2013)	Binary PSO	69	NO
Zhu et al. (2019)	MIQCP	68	NO
Wu et al. (2018)	Variable neighborhood search	87	NO
Mohammadi et al. (2016)	Binary imperialistic competition algorithm	81	NO
Proposed	MILP	79	YES

In this section, first of all, similar to previous studies, the best solution of 300 bus test system is selected and next the final result is compared with other similar papers in the literature (shown in Table 8).

As shown in Table 8, the proposed method has an acceptable number of PMUs in comparison with others. As shown, some papers like Wu et al. (2018), Mohammadi et al. (2016) suggested to install 87 or 81 PMUs. That is while, the proposed method has obtained 79 PMUs in parallel to maximizing the number of FLO lines to 224. In other words, if the proposed method was to minimize only the number of PMUs, the number of FLO lines would be much less than 224 (in the limit case, the FLO lines are equal to 145).

5.6. Main findings of the study

The case studies simulated in this section result in the following main findings:

- Adding the number of PMU channels as an optimization variable to the formulation results in a more affordable placement in the network. In this way, the busbars having more connections to other buses are chosen more than others.
- Considering fault location observability as an objective function makes the PMU devices be installed at the busbars connected to the lengthy lines. As a result, the PMU-based fault location algorithms would better distinguish the fault location at the lines experiencing the most probable faults.
- Using the ε-constraint method, the present method, the presented approach splits the multi-objective formulation to several single-oted approach splits the multi-objective formulation to several single-objective problems. Given these problems can each be modeled as a MILP problem to be solved via the famous CPLEX solver of GAMS, the resulted optimal Pareto front is a summation of several global optimal solutions. Therefore, the decision-maker's solution would be always global optimum.
- The formulation used for defining FLO index leads the PMU placement problem to install the PMUs at each end of the lines. That is, installing PMUs at both ends of lines has an extra value in formulation. In this way, the more accurate double-ended fault location algorithms can better find the location of probable faults.
- The FLO/PMU is only an index helping the decision-maker to find out which solution is more affordable with respect to its neighbors in the Pareto front; this index cannot eliminate

the decision-maker's role in selecting the best choice and it only help him have a better selection.

6. Conclusion

Two objective functions were presented in this paper to be used in PMU placement of a practical network. The former denoted the cost function of installing modular PMUs in the network. Given the PMU may have several current channels with respect to the number of lines connected to the bus, the cost function considered the impact of channels in the total cost of PMU placement. The latter was focused on the fault location capability of the network. To consider this fact, an objective function, named f_{FLO}, was presented and next utilized in the respected multi-objective formulation. Using a MILP-based model in GAMS software under CPLEX solver, the presented formulation was applied on the practical network of OMAN country. Solving the MILP program several times to reach the respected global solution, the simulation part presents the optimal Pareto front via the ε-constraint method. Given selecting the best choice from the obtained solutions may be confusing, an index named FLO/PMU was also presented to help the decision-maker select his/her choice from the Pareto optimal solutions. The selected solution, finally, was presented in detailed to illustrate how the trade-off between the cost function and FLO index changes the characteristics of the final solution. According to the importance of fault location observability during a practical application, considering the impact of noise in PMU data on the OPP results within a numerical observability formulation would be a good suggestion for a future work.

CRediT authorship contribution statement

Amer Al-Hinai: Methodology, Supervision, Writing – review & editing. **AliReza Karami-Horestani:** Methodology, Writing – review & editing. **Hassan Haes Alhelou:** Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge the support of the Sultan Qaboos University, the Ministry of Higher Education, Research and Innovation, Oman Electricity Transmission Company, and Occidental Oman. This paper is an outcome of the Block Funding program "Towards Integrated Monitoring System for Migration to Smart Grid in Oman (TIMSOman)" Project code (RC/RG-ENG/ECED/19/03).

References

Abdalla, O.H., Al-Hadi, H., Al-Riyami, H., 2009. Development of a digital model for oman electrical transmission main grid. In: 2009 International Conference on Advances in Computational Tools for Engineering Applications. IEEE, pp. 451–456.

Ahmed, M.M., Amjad, M., Qureshi, M.A., Imran, K., Haider, Z.M., Khan, M.O., 2022. A critical review of state-of-the-art optimal PMU placement techniques. Energies 15 (6), 2125.

Alexopoulos, T.A., Manousakis, N.M., Korres, G.N., 2016. Fault location observability using phasor measurements units via semidefinite programming. IEEE Access 4, 5187–5195.

Almasabi, S., Mitra, J., 2018. Multistage optimal PMU placement considering substation infrastructure. IEEE Trans. Ind. Appl. 54 (6), 6519–6528. http://dx.doi.org/10.1109/TIA.2018.2862401.

Andreoni, R., Macii, D., Brunelli, M., Petri, D., 2021. Tri-objective optimal PMU placement including accurate state estimation: The case of distribution systems. IEEE Access 9, 62102–62117.

Appasani, B., Jha, A.V., Swain, K., Cherukuri, M., Mohanta, D.K., 2022. Resiliency estimation of synchrophasor communication networks in a WAMS. Front. Energy Res. 350.

Arpanahi, M.K., Alhelou, H.H., Siano, P., 2019. A novel multiobjective OPP for power system small signal stability assessment considering WAMS uncertainties. IEEE Trans. Ind. Inform. 16 (5), 3039–3050.

Baba, M., et al., 2021. Optimization of phasor measurement unit placement using several proposed case factors for power network monitoring. Energies 14 (18), 5596.

Babu, R., Gupta, V.K., Subbaramaiah, K., 2022. An approach to unravel the optimal PMU placement problem for full observability of power network in view of contingencies. Int. J. Syst. Assur. Eng. Manag. 13 (3), 1170–1186.

Cao, B., et al., 2022. A multiobjective intelligent decision-making method for multistage placement of PMU in power grid enterprises. IEEE Trans. Ind. Inform. 1–9. http://dx.doi.org/10.1109/TII.2022.3215787.

Dixit, A., Chowdhury, A., Saini, P., 2022. A review on optimal placement of phasor measurement unit (PMU). Syst. Assur. 513–530.

Elimam, M., Isbeih, Y.J., Moursi, M.S.E., Elbassioni, K., Hosani, K.H.A., 2021. Novel optimal PMU placement approach based on the network parameters for enhanced system observability and wide area damping control capability. IEEE Trans. Power Syst. 36 (6), 5345–5358. http://dx.doi.org/10.1109/TPWRS.2021.3076576.

Galvez, C., Abur, A., 2022. Fault location in power networks using a sparse set of digital fault recorders. IEEE Trans. Smart Grid 13 (5), 3468–3480. http://dx.doi.org/10.1109/TSG.2022.3168904.

Hamedani Golshan, M.E., Hosseini Dolatabadi, S.H., Tabatabaei, S.M., 2018. Determining minimum number and optimal placement of PMUs for fault observability in one-terminal algorithms. IET Gener. Transm. Distrib. 12 (21), 5789–5797.

Hyacinth, L.R., Gomathi, V., 2021. Optimal pmu placement technique to maximize measurement redundancy based on closed neighbourhood search. Energies 14 (16), 4782.

Jha, A.V., Appasani, B., Ustun, T.S., 2022. Resiliency assessment methodology for synchrophasor communication networks in a smart grid cyber–physical system. Energy Rep. 8, 1108–1115.

Jia, K., Thomas, D.W.P., Sumner, M., 2013. A new double-ended fault-location scheme for utilization in integrated power systems. IEEE Trans. Power Deliv. 28 (2), 594–603. http://dx.doi.org/10.1109/TPWRD.2013.2238560.

Johnson, T., Moger, T., 2021. A critical review of methods for optimal placement of phasor measurement units. Int. Trans. Electr. Energy Syst. 31 (3), e12698. http://dx.doi.org/10.1002/2050-7038.12698.

Joshi, P.M., Verma, H.K., 2021. Synchrophasor measurement applications and optimal PMU placement: A review. Electr. Power Syst. Res. 199, 107428. http://dx.doi.org/10.1016/j.epsr.2021.107428.

Khajeh, K.G., Bashar, E., Rad, A.M., Gharehpetian, G.B., 2017. Integrated model considering effects of zero injection buses and conventional measurements on optimal PMU placement. IEEE Trans. Smart Grid 8 (2), 1006–1013. http://dx.doi.org/10.1109/TSG.2015.2461558.

Lin, W., Xia, B., Tang, Y., Zhang, T., Qi, Y., 2022. AoI research on PMU cloud side cooperative system of active distribution network. Energy Rep. 8, 677–686.

Lv, Q., Chen, Y., Yuan, K., Qin, L., Yu, X., Gui, H., 2022. Optimal sensor placement in distribution network based on super resolution network. Energy Rep. 8, 1050–1058.

Mazhari, S.M., Monsef, H., Lesani, H., Fereidunian, A., 2013. A multi-objective PMU placement method considering measurement redundancy and observability value under contingencies. IEEE Trans. Power Syst. 28 (3), 2136–2146. http://dx.doi.org/10.1109/TPWRS.2012.2234147.

Mishra, S.K., Cherukuri, M., 2020. Optimal placement of PMU: A brief review. In: Pradhan, G., Morris, S., Nayak, N. (Eds.), Advances in Electrical Control and Signal Systems. Springer Singapore, Singapore, pp. 529–540, 2020.

Mohammadi, M.B., Hooshmand, R.A., Fesharaki, F.H., 2016. A new approach for optimal placement of PMUs and their required communication infrastructure in order to minimize the cost of the WAMS. IEEE Trans. Smart Grid 7 (1), 84–93. http://dx.doi.org/10.1109/TSG.2015.2404855.

Naidu, O.D., Pradhan, A.K., 2021. Precise traveling wave-based transmission line fault location method using single-ended data. IEEE Trans. Ind. Inform. 17 (8), 5197–5207. http://dx.doi.org/10.1109/TII.2020.3027584.

Patel, C.D., Tailor, T.K., Shukla, S.K., Shah, S., Jani, S.N., 2022. Steiner tree-based design of communication infrastructure with co-optimizing the PMU placement for economical design of WAMS. IEEE Trans. Instrum. Meas. 71, 1–11. http://dx.doi.org/10.1109/TIM.2022.3179510.

Qi, J., Sun, K., Kang, W., 2015. Optimal PMU placement for power system dynamic state estimation by using empirical observability Gramian. IEEE Trans. Power Syst. 30 (4), 2041–2054. http://dx.doi.org/10.1109/TPWRS.2014.2356797.

Theodorakatos, N.P., 2019. Fault location observability using phasor measurement units in a power network through deterministic and stochastic algorithms. Electr. Power Compon. Syst. 47 (3), 212–229.

Wu, Z., et al., 2018. Optimal PMU placement considering load loss and relaying in distribution networks. IEEE Access 6, 33645–33653. http://dx.doi.org/10.1109/ACCESS.2018.2841891.

Zhang, M., Wu, Z., Yan, J., Lu, R., Guan, X., 2022. Attack-resilient optimal PMU placement via reinforcement learning guided tree search in smart grids. IEEE Trans. Inf. Forensics Secur. 17, 1919–1929. http://dx.doi.org/10.1109/TIFS.2022.3173728.

Zhu, X., Wen, M.H.F., Li, V.O.K., Leung, K.C., 2019. Optimal PMU-communication link placement for smart grid wide-area measurement systems. IEEE Trans. Smart Grid 10 (4), 4446–4456. http://dx.doi.org/10.1109/TSG.2018.2860622.